

Thesis Meeting 1

How Stable are Baselines: analysis on the Consistency of Reinforcement Learning Baselines



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Project overview

Setting:

- There is rapid growth in the number of baseline frameworks for Reinforcement Learning algorithms. These frameworks provide stable implementations of published algorithms, for easy use, convenience, reliability and consistency.

Problem:

- Due to this rapid growth, it is at question whether the same algorithms across different baseline frameworks perform consistent to each other and the original paper it was published in. Furthermore, across baselines, the relative ranking should therefore also be consistent.

Solution:

- Extensive literary search on baseline frameworks, algorithms, and environments to get a perspective of current reports
- Perform experiments on a fixed set of environments with algorithms across different frameworks to see whether performance is consistent.

Contribution:

- Filling a gap in RL literature by providing knowledge about algorithm comparisons across different baseline frameworks and algorithms, highlighting the consistency among algorithms between frameworks and relative ranking across frameworks.

Experimental questions:

- To what extent do results reported in original papers align with those produced by existing baseline framework implementations?
- To what extent can results obtained from different baseline frameworks be considered directly comparable?
- To what extent is the relative performance ranking of algorithms preserved across different baseline frameworks?

Timeline

Start date: 19 Feb

Current: 3 Mar

Projected first draft: 15 Aug

Projected final draft: 31 Aug

Month	Plan	Product
Feb 19	Finish documents (formal proposal, contract), start with Thesis.	Project Proposal, contract
Mar 11	Looking into the different frameworks that are available. I will start with a vast selection of frameworks, and look into the related work as to what has motivated this framework to exist and what other frameworks exist. Making such an overview gives good insight and makes comparing frameworks more accessible.	I want to have a list of frameworks (that have implemented PPO) and a list of environments (where PPO has been experimented with via these frameworks) ready by this point.
Mar 25/Apr 1	Having the list of environments and baseline frameworks ready, I want to see whether I can run one iteration of an environment of each framework on an environment, to see how the frameworks' architecture works, how logging results work, and whether the list of frameworks must be adjusted. In the meantime, gathering an overview of the literature of these baseline frameworks and their results outlines its consistency among experiments even more.	I want to have an overview of the baseline frameworks' implementations, and possible difficulties as to how to implement each experiment accordingly. A time estimation as to how long each experiment may take would also be great to have by this point, such that further planning may be more efficiently done. Furthermore, the list of experimental results using baselines may grow continuously.

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Apr 15	Set up the experiments on ALICE. I assume this will take some time and won't be as trivial, given the many versions and types of framework baselines. Given the ETA of the experiments, extending upon literary research is possible. However, up to this point we cannot decide yet whether to extend this to more algorithms, and thus additional literary research upon the reported performance on other algorithms is still 'redundant'.	Properly have everything set up on ALICE, and running. This includes having extensive loggers, using similar SLURM job attributes (such as CPUs/GPUs) for fair comparison. By this point, create a plan to extend this research to more overlapping algorithms from baselines,
Experiment 1	After the experiment with PPO applied on a fixed set of environments with implementations of PPO from several baselines, we can create a more thorough hypothesis and plan continuing onwards. This includes, but is not limited to, finding out what causes these inconsistencies between implementations of PPO, extending our set of environments to be tested on, or finding potential other algorithms to look into RQ3	A plan of the aforementioned proportions on how to continue with this research. This plan, similar to a plan as this one, would give more insight into the deadlines of experiments to validate our research questions, the time spent on literary research of other algorithms, and whether we should change some attributes of our experiments.

Completed steps

1. Finalized bureaucracy in order to start
2. Set up my own organization (GitHub, ALICE, Overleaf, including Todos and Question lists)
3. Gathered a list of 8* baseline frameworks and made an overview on what algorithms were implemented
4. Get familiar with PPO
5. Get a list of performance reports by the aforementioned baseline frameworks, along with experimental details (as far as possible to retrieve)
6. Understand the motivation for certain frameworks to be implemented
7. Make a comparison (as far as possible) with the current reported performances on environments to find outliers (-> candidate environments)

Status

1. Retrieve a list of popular environments in baseline frameworks, and check the repository they come from
 - a. Status: in progress
 - b. Issue: scale

Topic to discuss (1/3)

Observation/problem:

- The list of baseline frameworks I currently have is, so to say shallow, sometimes poorly documented and/or no longer maintained

Hypotheses/explanation:

- Apart from Stable Baselines 3, CleanRL and TorchRL, I have initially picked an arbitrary set of baseline frameworks I came across. Although still mentioned throughout related work sections, some of these were outdated.

Possible verification:

- After looking into benchmarking results, some were unavailable, shallow, or just not well documented. Due to its age, some experiments also feel outdated, given the versions some environments were tested on.

Possible solutions:

- Find out the contribution to the relevancy of our research for these baseline frameworks
- Consider using other baseline frameworks instead.

Topic to discuss (2/3)

Observation/problem:

- From the few baseline frameworks I have looked into, it varies a lot how much benchmarking is done and how much experimental details are reported.
- This leads to reading learning graphs to get a glimpse of knowledge (though the differences at which baseline frameworks report their results also varies, so it is not so out of line after all)

Hypotheses/explanation:

- Due to the age of a framework, resources may not have been available for large-scale testing
- Frameworks were used in specific settings, not having the need for large-scale testing
- Forgotten/not reported due to other reasons

Possible verification:

- I could look into papers having used these baselines to see what they have reported with this algorithm, which is a step deeper as to what I am doing now.

Possible solutions:

- Using other frameworks for, in the end, better comparisons
- Dealing with the fact that there are sparse reportations done on the performance
- Look deeper into the literature for a certain baseline to get a (concise) list of performance reports

Topic to discuss (3/3)

Observation/problem:

- The number of environments vary by a lot. Not only in terms of task type, but also in version (v1,v2,etc.), the use of framestacking (in ALE environments), the use of different repositories (e.g. Atari Envpool), etc.
- The list I currently have obtained give an overview of specific overshadowing collections of environments, but in specificity it is very loosely defined which algorithms may be beneficial to use for experimentation

Hypotheses/explanation:

- Environments may get updated throughout time, tweaking small settings. Furthermore, I hypothesize that some repositories may optimize due to faster emulators, parallel processing options or support distributed learning.

Possible verification:

- Find the attributes of the repositories

Possible solutions:

- Look into what made each version different from each other, for which I assume are insignificant differences
- This will allow to state whether environments with different versions may be compared.

To Do (next meeting)

- Finalize the baseline framework list
- Update the listings (available algorithms, framework properties, PPO performance reports) accordingly,
- Get a list of candidate environments for experimentation